FIITJEE REVIEW TEST-01

(Batches: 921 Lot)

IIT – JEE, 2021

Time: 1 Hour **Maximum Marks: 59**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-04)** contains **04 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (05-09)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-A – (10-13)** contains **02 paragraphs**. Based on each paragraph, there are 2 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
5. **Part-C – (14-18)** contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks	:	+ 1	For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
Zero Marks	:	0	If none of the bubbles is darkened.
Negative Marks:		– 2	In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part A (4),** you will be **awarded +3 marks** for the correct answer and **-1 marks** for wrong answer.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

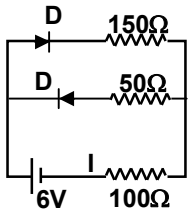
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SECTION : PHYSICS

PART – A: (Single Correct Answer Type)

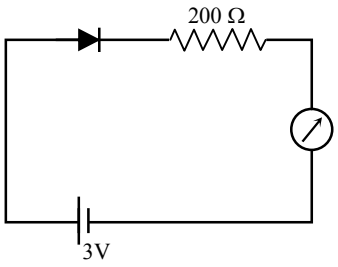
*This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. The circuit shown in the figure contains two diodes each with a forward resistance of 50 ohms and with infinite backward resistance. If the battery voltage is 6 V, the current through the 100 ohm resistance (in Amperes) is



(A) zero (B) 0.02
(C) 0.03 (D) 0.036

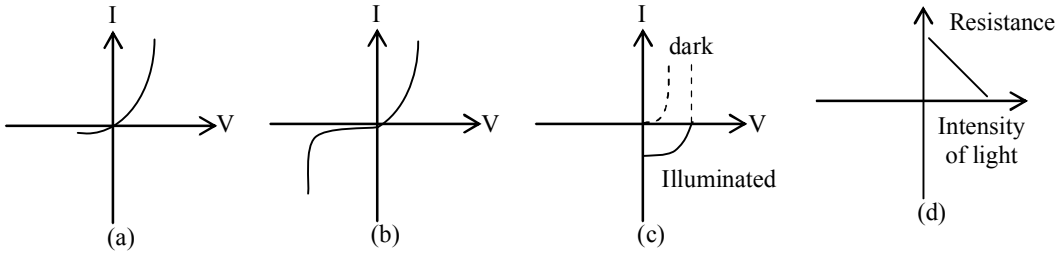
2. The reading of the ammeter for a silicon diode in the given circuit is :



(A) 13.5 mA (B) 0
(C) 15 mA (D) 11.5 mA

3. In n-type silicon, which of the following statement is true.
(A) electrons are majority carriers and trivalent atoms are the dopants.
(B) Electrons are minority carriers and pentavalent atoms are the dopants.
(C) Holes are minority carriers and pentavalent atoms are the dopants.
(D) Holes are majority carriers and trivalent atoms are dopants.

4. Identify the semiconductor devices whose characteristics are given below, in the order (1), (2), (3), (4) :



(A) Zener diode, Simple diode, Light dependent resistance, Solar cell
(B) Solar cell, Light dependent resistance, Zener diode, Simple diode
(C) Zener diode, Solar cell, Simple diode, Light dependent resistance
(D) Simple diode, Zener diode, Solar cell, Light dependent resistance

Space for rough work

PART – A(II) (Multiple Correct Choice Type)

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

5. A particle of mass m moves on the x-axis as follows: it starts from rest at t=0 from the point x = 0, and comes to rest at t=1 at the point x=1. No other information is available about its motion at intermediate times (0<t<1). If α denotes the instantaneous acceleration of the particle, then
   (A) α cannot remain positive for all t in the interval $0 \le t \le 1$
   (B) $|\alpha|$ cannot exceed 2 at any point in its path
   (C) $|\alpha|$ must be ≥ 4 at some point or points in its path
   (D) α must change sign during the motion but no other assertion can be made with the information given.

6. Man A sitting in a car moving with 54 km/hr observes man B in front of car crossing perpendicularly the road of width 15 m in 3 s.
   (A) Speed of man B is $5\sqrt{10}$ m/s
   (B) Speed of man B is 5 ms^{-1}
   (C) Actual direction of motion of B is at an angle of $\tan^{-1}\left(\frac{1}{3}\right)$ with direction of motion of car.
   (D) Actual direction of motion of B is at an angle of $\tan^{-1}(3)$ with direction opposite to the direction of motion of car.

7. A simple pendulum of length L and mass M is oscillating in a plane about a vertical line between angular limits - ϕ and +ϕ. For an angular displacement θ ($|\theta| < \phi$) , the tension in the string and the velocity of the bob are T and V respectively. The following relations hold good under the above conditions
   (A) $T\cos\theta = Mg$
   (B) $T - Mg\cos\theta = MV^2/L$
   (C) The magnitude of the tangential acceleration of the bob $|a_T| = g\sin\theta$
   (D) $T = Mg\cos\theta$

8. For a curved track of radius R, banked at angle θ
   (A) a vehicle moving with a speed $V_0 = \sqrt{Rg\tan\theta}$ is able to negotiate the curve without calling friction into play at all.
   (B) a vehicle moving with any speed $V > V_0$ is able to negotiate the curve, with friction called into play.
   (C) a vehicle moving with any speed $V < V_0$ must also have the force of friction into play.
   (D) the minimum value of the angle of blanking for a vehicle parked on the blanked road can stay there without slipping, is given by $\theta = \tan^{-1}\mu_s$ (μ_s = coefficient of static friction)

9. A person applies a constant force $\vec{F}$ on a particle of mass m and finds that the particle moves in a circle of radius r with a uniform speed v as seen (in the plane of motion) from an inertial frame of reference. Select the correct statement.
   (A) This is not possible
   (B) There are other forces on the particle
   (C) The resultant of the other forces is mv^2/r towards the centre
   (D) The resultant of the other forces varies in magnitude as well as in direction

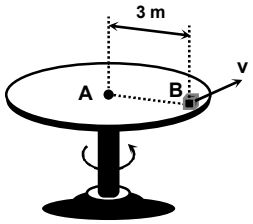
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PART – A(III): (Paragraph Type)

*This section contains **2 paragraphs.** Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 10 to 11

A small block B is supported by a turntable which starting from rest is rotated in such a way that the block undergoes a constant tangential acceleration $a_t = 4\ m/s^2$. The distance between axis o rotation and the block is 3 m. Knowing that the coefficient of static friction between the block and the turntable is 0.50

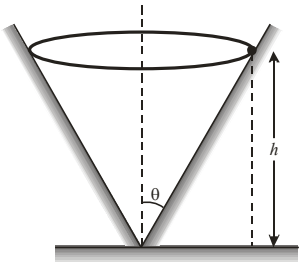


10. How long it will take for the block to start slipping on the turntable, is
(A) 2 sec (B) 0.75 sec
(C) 0.5 sec (D) None of these.

11. The speed v of the block at the instant when slipping will start, is
(A) 6 m/s (B) 5 m/s
(C) $3\ m/s^2$ (D) None of these

Paragraph for Question Nos. 12 to 13

A particle of mass 'm' is moving in horizontal circle inside a smooth inverted vertical cone above height 'h' from apex. Half angle of cone is 'θ' then give the answer of following questions.



12. Normal force by surface of cone is :
(A) mg cos θ (B) mg sec θ
(C) mg sin θ (D) mg cosec θ

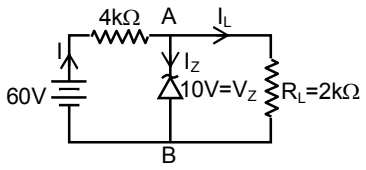
13. Speed of particle to move in horizontal circle if h increases keeping θ constant:
(A) will remain same (B) will decrease
(C) will increase (D) first increase then decrease

Space for rough work

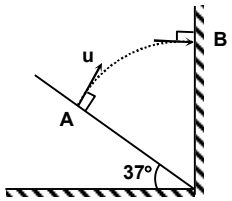
PART-B
(Numerical Answer Type)

This section contains **05** questions. The answer to each question is a **single-digit integer**, ranging from **0 to 9**.

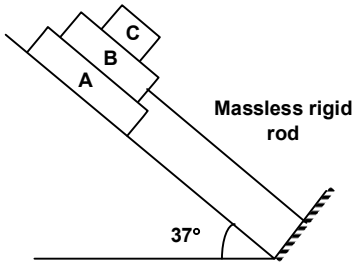
14. A Zener diode is connected to a battery and a load as shown below: If the currents in 4kΩ is $\frac{25}{\lambda}$. Then find the value of λ.



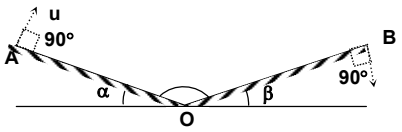
15. A particle is projected from point A perpendicular to inclined plane with a velocity 50 m/s as shown in the figure. Particle strikes a vertical plane perpendicularly at point B. Time taken by particle in going from point A to point B is



16. In the shown figure m_A, m_B and m_C are the masses of the three blocks, the incline is frictionless m_A = 5 kg, m_B = 10 kg, m_C = 2kg. coefficient of friction between A and B is 0.5 and between B and C is 0.1. The frictional force on block A is 6K Newton. Find the value of K. (Take g = 10 m/s^2)



17. Two inclined planes OA and OB of inclinations to the horizontal are α and β, each equal to 30° as shown in the figure. A particle is projected at an angle of 90° with plane OA from point A and its strikes the plane OB at point B normally. Then find the speed of projection in m/s. (given that OA = OB = 20 cm and g = 10 m/s^2)



18. Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\hat{i}$ and $\vec{B} = a\left(\cos\omega t\ \hat{i} + \sin\omega t\ \hat{j}\right)$, where a is a constant and $\omega = \pi/6$ rad s^{-1}. If $\left|\vec{A}+\vec{B}\right| = \sqrt{3}\left|\vec{A}-\vec{B}\right|$ at time t = τ for the first time, the value of τ, in seconds, is _____.

Space for rough work

FIITJEE REVIEW TEST-01

(Batches: 921 Lot)

IIT – JEE, 2021

ANSWERS

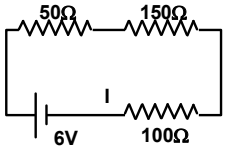
PHYSICS

1.	**B**	**2.**	**D**	**3.**	**C**	**4.**	**D**
5.	**A, C**	**6.**	**A, C**	**7.**	**B, C**	**8.**	**A, B, C, D**
9.	**B, D**	**10.**	**B**	**11.**	**C**	**12.**	**D**
13.	**C**	**14.**	**2**	**15.**	**4**	**16.**	**5**
17.	**2**	**18.**	**2**				

SOLUTIONS

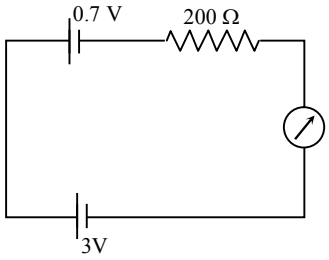
PHYSICS

1. As D_2 is reverse biased the given circuit can be redrawn as



Hence the current I through 100Ω resistor = $\frac{6}{300} = 0.02A$

2. $I = \frac{3-0.7}{200} A = 11.5$ mA



5. At initial and final point, body is at rest, this is possible only when its acceleration changes sign during the motion.

Let its minimum acceleration is α then $\frac{1}{2} = \frac{1}{2}\alpha\left(\frac{1}{2}\right)^2 \Rightarrow \alpha = 4$

Hence $|\alpha| \geq 4$

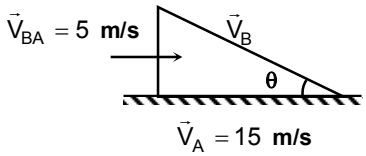
6. $\vec{V}_B = \vec{V}_{BA} + \vec{V}_A$

$\left|\vec{V}_B\right| = \left|\vec{V}_{BA} + \vec{V}_A\right|$

$= \sqrt{V_{BA}^2 + V_A^2} = \sqrt{5^2 + 15^2} = 5\sqrt{10}$ m/s

$\tan\theta = \frac{5}{15} = \frac{1}{3}$

$\theta = \tan^{-1}\left(\frac{1}{3}\right)$



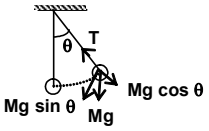
7. $T - Mg\cos\theta = \frac{Mv^2}{L}$

and $|a_T| = g\sin\theta$

At maximum angular displacement

$T - Mg\cos\phi = 0$

so $T = Mg\cos\phi$



9. The resultant of all forces may provide a constant magnitude of acceleration mv^2/r towards the centre.

10. $T = v/a_t$

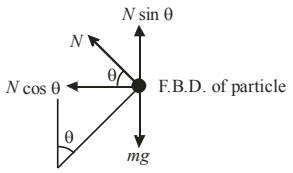
11. $\frac{v^2}{R} = a_r$

$a_r = \sqrt{a_{net}^2 - a_t^2} = \sqrt{25-16} = 3$ m/s^2

13. $N\sin\theta = mg \Rightarrow N\cos\theta = \frac{mv^2}{R}$

$\tan\theta = \frac{gR}{v^2} = v = \sqrt{gR\cot\theta}$

$v = \sqrt{gh}$ as h increases v increases.



15. $v_y = u_y + a_y t$

$0 = u\cos 37° - gt$

$t = \frac{u\cos 37°}{g} = \frac{50}{10} \times \frac{4}{5} = 4 \text{ sec}$

16. $m_{Ag} \sin\theta = 5 \times 10 \times \sin 37° = 5 \times 10 \times \frac{3}{5} = 30 \text{ N}$

$f_{max} = \mu_{AB}(m_C + m_B)g\cos\theta = 48 \text{ N}$

hence f = 30 N

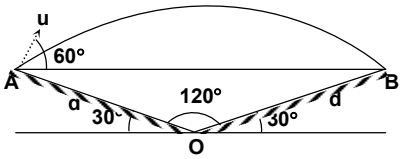
17. Since $\alpha = \beta$

AO = BO = 20 cm

$AB = \sqrt{d^2 + d^2 - 2d \cdot d\cos 120°}$

$AB = \sqrt{3}d$

Now AB is the range of the projectile from A to B.



$\sqrt{3}d = \frac{u^2 \sin 2\theta}{g}$

$\Rightarrow \sqrt{3}d = \frac{u^2\sqrt{3}}{2g}$

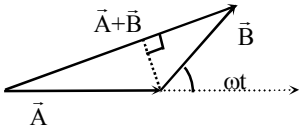
Putting d = 0.2 m

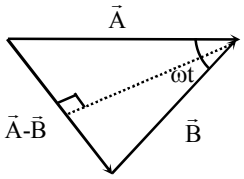
u = 2 m/s

18. $\therefore 2a\cos\frac{\omega t}{2} = \sqrt{3}(2a)\sin\frac{\omega t}{2}$

$\therefore \tan\frac{\omega t}{2} = \frac{1}{\sqrt{3}}$

for the first time $\Rightarrow \frac{\pi t}{12} = \frac{\pi}{6} \Rightarrow t = 2\text{sec}$





FIITJEE REVIEW MOCK TEST-01

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

Time: 1 Hour **Maximum Marks: 59**

INSTRUCTIONS

A. Question Paper Format

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2. **Part-A – (01-04)** contains **04 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
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4. **Part-A – (10-13)** contains **02 paragraphs**. Based on each paragraph, there are 2 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
5. **Part-C – (14-18)** contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

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Partial Marks : + 1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

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For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part A (4),** you will be **awarded +3 marks** for the correct answer and **-1 marks** for wrong answer.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **.Session :**. . . . . **Date:**. . . . . .

PART – A(I): (Single Correct Answer Type)

*This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. Imagine that in any atom about 50% of the space is occupied by the atomic nucleus. If a silver foil is bombarded with α-particles, majority of the α-particles would:
   (A) get scattered (B) be absorbed by the nuclei
   (C) pass through the foil undetected (D) get converted into photons

2. Which electronic transition in a hydrogen atom releases the greatest amount of energy?
   (A) $n = 3 \rightarrow n = 2$ (B) $n = 5 \rightarrow n = 3$
   (C) $n = 6 \rightarrow n = 5$ (D) $n = 3 \rightarrow n = 6$

3. Among the following, the molecule with the highest dipole moment is
   (A) NH_3 (B) NF_3
   (C) CH_4 (D) CCl_4

4. Unpaired electrons in $[Co(H_2O)_6]^{2+}$ are
   (A) 5 (B) 2
   (C) 3 (D) 4

PART – A(II): (Multi Correct Answer Type)

*This section contains **05 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

5. NH_3 and BF_3 form adduct readily
   (A) Hybridization of NH_3 remains same.
   (B) through co–ordinate bond between B and N
   (C) Hybridization of NH_3 changes from sp^3 to sp^2.
   (D) Hybridization of B changes from sp^2 to sp^3.

6. Monochromatic light of wavelength λ strikes a metal surface with intensity *'a'* emitting 'b' electrons per second with 'c' as the maximum kinetic energy of electrons. Which of the following options will happen if 'a' is halved without changing wavelength and area of exposure?
   (A) Photo emission will stop.
   (B) New maximum kinetic energy will be same as 'c'.
   (C) Number of photoelectrons emitted per second will be less than *'b'.*
   (D) Rate at which charge will be acquired on metal will decrease.

Space for rough work

7 Bohr model can be applied on :
(A) H atom (B) He^{+} ion
(C) H^{-} ion (D) Li^{2+} ion

8. The isotopes contain the same number of
(A) neutrons (B) protons
(C) protons + neutrons (D) elecrons

9. The properties which is common to both groups 1 and 17 elements in the periodic table are
(A) electropositive character increases down the groups
(B) reactivity decreases from top to bottom in these groups
(C) atomic radii increases as the atomic number increases
(D) electronegativity decreases on moving down a group

PART – A(III): (Paragraph Type)

*This section contains **2 paragraphs.** Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 10 to 11

Ionic bond is defined as the electrostatic force of attraction holding the oppositely charged ions. Ionic compounds are mostly crystalline solids having high melting and boiling points, electrical conductivity in molten state, solubility in water etc. Covalent bond is defined as the force which binds atoms of same or different elements by mutual sharing of electrons in a covalent bond. Covalent compounds are solids, liquids or gases. They are low melting and boiling point compounds. They are more soluble in non–polar solvents.

10. The valence electrons not involved in formation of covalent bonds are called
(A) non–bonding electrons (B) lone pairs
(C) unshared pairs (D) all of these

11. Which of the following gives a white precipitate with $AgNO_3$?
(A) NaCl (B) CCl_4
(C) CO_2 (D) $CHCl_3$

Space for rough work

Paragraph for Question Nos. 12 to 13

In an atom when an electron jumps from higher energy level to lower energy level it emits energy in form of electromagnetic radiations. When these electromagnetic radiations are passed through a prism and received on a photographic film some lines are observed on that film and those lines are called spectral lines.

For hydrogen like species, when jump takes from any excited state to ground state (n = 1), line produced in that case is called a Lyman series line.

If transition occurs from 3rd or above levels to second level then corresponding lines produced are called Balmer lines. Similarly, for next level it is called Paschen series line.

12. The wavelength of photon corresponding to second least energy of Lyman series in H-atom is :
(A) 102 nm (B) 121 nm
(C) 91 nm (D) 486 nm

13. If there are 3 atoms of a hydrogen like species one in 2nd, one in 3rd and one in 4th excited state, then how many maximum total different Balmer and Paschen lines can be produced?
(A) 2 (B) 3
(C) 4 (D) 5

PART-C (Numerical Based Type)

This section contains 5 questions. The answer to each question is a single-digit integer, ranging from 0 to 9.

14. How many unpaired electrons are present in N_2^+?

15. An element with atomic number 20 will be placed in …….. period of the Periodic Table

16. The maximum number of valency electrons possible for atoms in the second period of the Periodic Table is

17. How many of the following relations are correct regarding the given property?
(1) C < N > O < F: First ionization enthalpy
(2) Cl > F > Br: First electronaffinity
(3) O > S > Se: Electronegativity
(4) Li > Be > B: Second ionization enthalpy

18. In Na^+, Mg^{+2}, Al^{+3}, O^{-2}, F^-, Cl^- atoms, how many atoms are having less radius as compare to Al^{+3}?.

Space for rough work

FIITJEE REVIEW TEST-01

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

ANSWER KEYS

1. **A**
2. **A**
3. **A**
4. **C**
5. **A, B, D**
6. **B, C, D**
7. **A, B, D**
8. **B, D**
9. **A, C, D**
10. **D**
11. **A**
12. **A**
13. **B**
14. **1**
15. **4**
16. **8**
17. **3**
18. **0**

FIITJEE REVIEW TEST-01

(Batches: 921 Lot)

IIT – JEE, 2021

Time: 1 Hour **Maximum Marks: 59**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-04)** contains **04 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (05-09)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-A – (10-13)** contains **02 paragraphs**. Based on each paragraph, there are 2 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
5. **Part-C – (14-18)** contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks	:	+ 1	For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
Zero Marks	:	0	If none of the bubbles is darkened.
Negative Marks:		– 2	In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part A (4),** you will be **awarded +3 marks** for the correct answer and **-1 marks** for wrong answer.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . .**Session :**. . . . . **Date:**. . . . . .

SECTION : MATHEMATICS

PART – A: (Single Correct Answer Type)

*This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. The distance between the two lines represented by the equation $9x^2 – 24xy + 16y^2 – 12x + 16y – 12 = 0$ is
   (A) 8/5 (B) 6/5
   (C) 11/5 (D) none of these

2. Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, then 2r is equal to
   (A) $\sqrt{PQ \cdot RS}$ (B) $\frac{PQ + RS}{2}$
   (C) $\frac{2PQ \cdot RS}{PQ + RS}$ (D) $\frac{\sqrt{PQ^2 + RS^2}}{2}$

3. The locus of the point of intersection of the normals at the end of the focal chord of the parabola $y^2 = 4ax$ is
   (A) $y^2 = 4ax$ (B) $x^2 = 4ay$
   (C) $x^2 = a(y – 3a)$ (D) $y^2 = a(x – 3a)$

4. The locus of the orthocenter of the triangle formed by the lines $(1 + p)x – py + p(1 + p) = 0$, $(1 + q)x – qy + q(1 + q) = 0$ and $y = 0$, were $p \neq q$, is
   (A) a hyperbola (B) a parabola
   (C) an ellipse (D) a straight line.

Space for rough work

PART – A: (Multi Correct Answer Type)

*This section contains **05 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

5. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$, then
   (A) radius of S is 8 (B) radius of S is 7
   (C) centre of S is (–7, 1) (D) centre of S is (–8, 1)

6. If the normal to the parabola $y^2 = 4ax$ at point t_1 cuts the parabola again at point t_2 then possible values of t_2^2 is
   (A) 8 (B) 9
   (C) 7 (D) 6

7. Let 'L' be a normal to the parabola $y^2 = 4x$. If L passes through the point (9, 6) then 'L' is given by
   (A) $y - x + 3 = 0$ (B) $y + 3x - 33 = 0$
   (C) $y + x - 15 = 0$ (D) $y - 2x + 12 = 0$

8. The equations of the tangents drawn from the origin to the circle $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are
   (A) $x = 0$ (B) $y = 0$
   (C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$

9. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is equal to
   (A) 1 (B) 2
   (C) –1/2 (D) – 1

Space for rough work

PART – A: (Paragraph Type)

*This section contains **2 paragraphs.** Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

Paragraph for Question Nos. 10 to 11

Let PQ be a focal chord of the parabola $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a$, $a > 0$

10. The length of chord PQ is
(A) 7a (B) 5a
(C) 2a (D) 3a

11. If chord PQ subtends an angle 'θ' at the vertex of $y^2 = 4ax$ then $\tan\theta$ =
(A) $\frac{2\sqrt{7}}{3}$ (B) $-\frac{2\sqrt{7}}{3}$
(C) $\frac{2\sqrt{5}}{3}$ (D) $-\frac{2\sqrt{5}}{3}$

Paragraph for Question Nos. 12 to 13

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P\left(\sqrt{3}, 1\right)$. A straight line L, perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$

12. A possible equation of L is
(A) $x - \sqrt{3}y = 1$ (B) $x + \sqrt{3}y = 1$
(C) $x - \sqrt{3}y = -1$ (D) $x + \sqrt{3}y = 5$

13. A common tangent of the two circles is
(A) $x = 4$ (B) $y = 2$
(C) $x + \sqrt{3}y = 4$ (D) $x + 2\sqrt{2}y = 6$

Space for rough work

PART – C: (Integer Answer Type)

*This section contains **05 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.*

14. The area enclosed by the lines represented by $|x - 1| + |y - 2| = 1$ is equal to _____

15. The vertices of a triangle are A(–1, –7), B(5, 1) and C(1, 4). The equation of the bisector of $\angle ABC$ is $ax + by + 2 = 0$, then the value $|a + b|$ is equal to _____

16. If a circle passes through the points of intersection of the co-ordinate axes with the lines $\lambda x - y + 1 = 0$ and $x - 2y + 3 = 0$, then the value of λ is _____

17. If $x + y = k$ is normal to $y^2 = 12x$ then value of k is _____

18. Let 'S' be the focus of the parabola $y^2 = 8x$ and PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is _____

Space for rough work

FIITJEE REVIEW TEST

MATHEMATICS

ANSWERS

MATHEMATICS (SECTION –III)

PART – A

1. **A**	2. **A**	3. **D**	4. **D**
5. **B, C**	6. **A, B**	7. **A, B, D**	8. **A, C**
9. **A, D**	10. **B**	11. **D**	12. **A**
13. **D**			

PART – C

14. **2**	15. **6**	16. **2**	17. **9**
18. **4**			

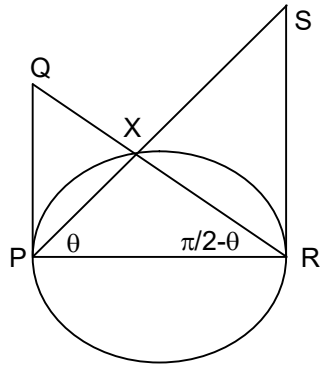
HINTS & SOLUTIONS

MATHEMATICS (SECTION –III)

PART – A

1. **A**
Factories and apply perpendicular distance formula.

2. **A**
In this diagram take $\tan\theta$ and $\cot\theta$, we get $\frac{PQ}{PR}\cdot\frac{RS}{PR}=1$
$\Rightarrow \quad 2r=\sqrt{PQ\cdot RS}$



3. **D**
$t_1t_2 = -1$
$h = 2a + a(t_1^2 + t_2^2 + t_1t_2)$
$k = -at_1t_2(t_1 + t_2)$
on simplication we get
$y^2 = a(x - 3a)$

4. **D**
On solving we get
$h = pq, k = -pq$ hence
locus is $x + y = 0$

5. **B, C**
Radical axis is $x + 7 = 0$
Centre of required circle $(-7, k)$ radius = length of the tangent from C to the any one of the given circle
$\Rightarrow \quad k = 1$ and $r = 7$.

6. **A, B**
$$t_2 = -t_1 - \frac{2}{t_1}$$
$\Rightarrow \quad t_1^2 + t_1t_2 + 2 = 0 \Rightarrow D \geq 0$
$\Rightarrow \quad t_2^2 \geq 8$

7. **A, B, D**
Equation of normal $y = mx - 2m - m^3$, it passes through (9, 6)
$\Rightarrow \quad m = 1, 2, -3$

8. **A, C**
$y = mx$ is tangent to the circle $x^2 + y^2 - 2rx - 2hy + h^2 = 0$
Now, the length of perpendicular from centre (r, h) on tangent = radius
$\Rightarrow \quad m = \infty, \frac{h^2 - r^2}{2hr}$

$\Rightarrow$ $x = 0$ and $(h^2 - r^2)x - 2rhy = 0$

9. **A, D**

Lines take the form $(y - mx)(my + x) = 0$

$\Rightarrow$ $y = mx$ or $y = (-1/m)x$

Bisectors of $xy = 0$ are $y = x$

And $y = -x \Rightarrow m = 1, -1$

10. **B**

11. **D**

$P(at^2, 2at)$ $Q(a/t^2, -2a/t)$ point of intersection of tangents at P and Q is $\left(-a, a\left(t - \frac{1}{t}\right)\right)$ this lies on line

$y = 2x + a$

$\Rightarrow$ $\left(t + \frac{1}{t}\right)^2 = 5 \Rightarrow PQ = 5a$

$$\tan\theta = \left|\frac{{}^{m}O_P - {}^{m}O_Q}{1 + {}^{m}O_P \cdot {}^{m}O_Q}\right| = -\frac{2\sqrt{5}}{3}$$

12. **A**

13. **D**

Equation of tangent $P\left(\sqrt{3}, 1\right)$ is $\sqrt{3}x + y = 4$

Equation of tangents with slope $\frac{1}{\sqrt{3}}$ to $(x-3)^2 + y^2 = 1$ will be

$$y = \frac{1}{\sqrt{3}}(x-3) \pm 1 \cdot \sqrt{1 + \frac{1}{3}}$$

$\Rightarrow$ $\sqrt{3}y = x - 1$ or $\sqrt{3}y = x - 5$

The point of intersection of direct common tangents is (6, 0) equation of common tangent

$y - 0 = m(x - 6)$ as it touches

$x^2 + y^2 = 4 \Rightarrow m = \pm\frac{1}{2\sqrt{2}}$

$\Rightarrow$ $y = -\frac{1}{2\sqrt{2}}(x - 6)$.

PART – C

14. **2**

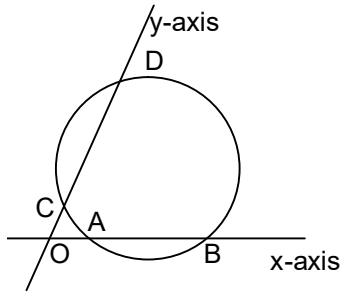
Lines represent a square with side length $\sqrt{2}$

15. **6**

Equation of angle bisector comes $x - 7y + 2 = 0$

16. **2**

Apply $OA \times OB$
$= OC \times OD$
$\Rightarrow \quad \lambda = 2$



17. **9**
On comparing with the normal equation, we get k = 9.

18. **4**
On solving we get P(0, 0), Q(2, 4) and S(2, 0)
$\Rightarrow$ area = 4 sq. units.